PDP-8/IXS Simulator User's Guide[1]

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Version 2.0

Summary

PDP-8/IXS is a Microsoft Windows 10/11 desktop application (C# WinForms) that implements a complete simulation of a Digital Equipment Corporation (DEC) PDP-8/I computer, including:

- Full instruction set
- Interrupts
- Single- and three-cycle data break
- Extended arithmetic element (EAE) KE8/I
- Memory extension control MC8/I and 8 memory fields MM8/I
- RK05 disk drives, 4 units
- DECtape (not yet implemented)
- ASR38/VT100 console terminal (limited escape sequences)
- AF01A analog to digital converter (two channels supported)
- Front panel with fully functioning lights and switches
- Actual execution speed can match the PDP-8/I at 1.5 μs per cycle.

Every attempt has been made to faithfully simulate the machine as documented by DEC. In addition, PDP-8/IXS emulates custom non-DEC hardware that existed on the MIT Weather Radar Research Project's PDP-8/I in the 1970's, some of which came from the Hurricane Research Project (HRP) of the US Air Force and some added by MIT.

- Index register (MIT)
- Priority interrupt controller (HRP)
- Supervisor call instructions (MIT)
- Real time 8 Hz clock (HRP)
- Interval timer with programmable intervals (MIT)
- Second ASR38/VT100 terminal (MIT)
- Tektronix 611 storage scope (HRP)
- Radar interfaces for two weather radar sets—digital sweep integrator, azimuth and elevation, antenna servo (HRP & MIT, currently partial emulation)
- Various displays and switches (HRP)

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PDP-8/IXS also has

- Simulated hardware breakpoint, using switch register to break on read or write of a specific address or data word.
- Event recorder for advanced code analysis and debugging
- Standard Windows File menu for creating, opening, and saving a `.pdp8` document that records the complete state of the machine.

Goals

- General purpose PDP-8/I simulation
- Running OS/8
- Running E8, the OS/8 Emacs-like editor
- Running the RADAR timesharing system developed at MIT
- Something for a hard-core nerd to do in retirement

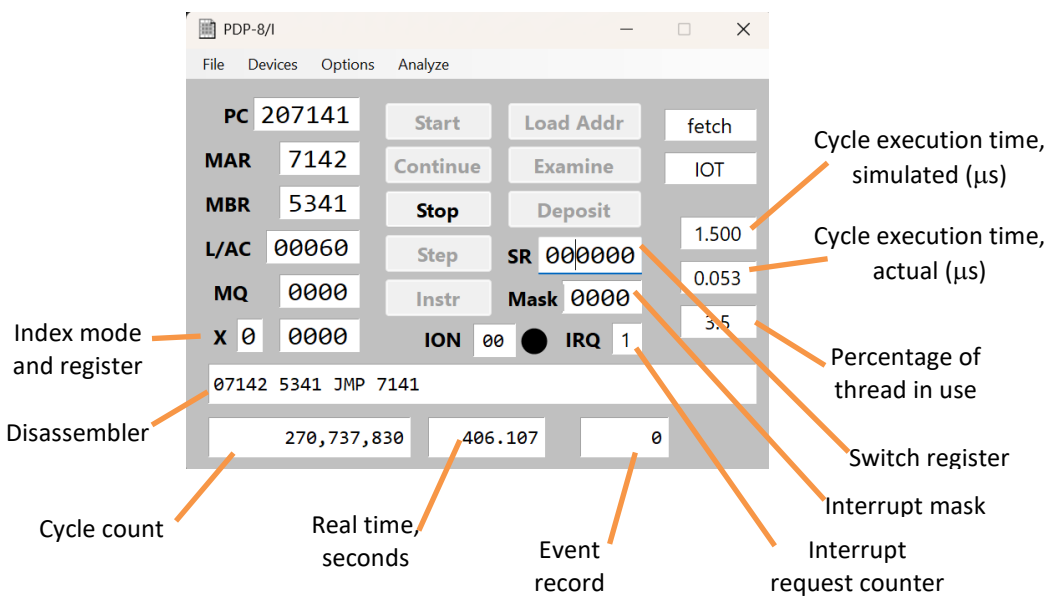
Apologies to the Linux/Mac communities. I'm playing the cards my engineering life has been delt.

Installation

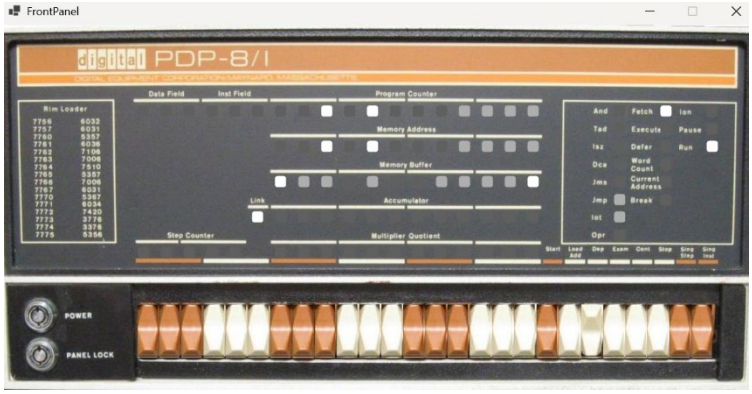
If you have Visual Studio 2022 you can clone the repository and build and run the C# project. You can do as you please with the source code subject to the license in the repository. If you have Windows 10 or 11 you can download the latest release from the repository, a standalone desktop app that has no external dependencies. Just unzip the file and run `PDP-8.exe`.

Front Panels

The main window is a modern GUI version of the PDP-8/I front panel, with some additions:



If you select Options→Show Front Panel you'll get this sizable window.



The lights are an accurate emulation of incandescent bulbs. All switches are functional and historically accurate. You can click or swipe the switches with a mouse or touchscreen. The switches can be set from either front panel and any changes in one are immediately reflected in the other.

Single Stepping and Interrupt Suppression

The following is historically accurate to the best of my knowledge except where noted.

A PDP-8 that is halted is actually quite active. All data break cycles are serviced, so RK05 and DECtape transfers are unaffected by halt. If interrupts are enabled and not suppressed, any interrupt that is not masked off will resume instruction execution to service the interrupt and then continue running. The HLT instruction is useless for debugging when interrupts are enabled.

Interrupts can be suppressed for three reasons:

- When ION is executed, suppress for one instruction to allow interrupt return
- When CIF is executed, suppress until next JMP or JMS
- For breakpoints (not historical) and single instruction stepping.

On a PDP-8/I, when the Single Instr switch was set the CPU halted with interrupts suppressed, and the Continue switch allowed single instruction stepping without interrupt interference. PDP-8/IXS does this, and in addition halts with interrupts suppressed for Sing Step (step one cycle), Stop, and when a breakpoint is hit.

Front Panel State Displayed

When the PDP-8 halts the front panels show the state of the CPU at the end of the cycle (which may or may not be the end of a multi-cycle instruction). The PC and disassembler show the next instruction to be executed. The MBR may or may not show the contents of memory at the MAR location, depending on the cycle type.

PDP-8 Documents

A PDP8 document is a file with a `.pdp8` extension that records the complete state of the machine, including all CPU registers and states, all I/O devices, the state of the console switches, the contents of memory, and the size and position of all PDP-8/IXS windows. The contents of the RK05 disks are currently stored in separate files.

The File menu contains the usual Windows commands (New, Open, Save, Save As) for working with documents. These commands can be invoked at any time, whether the PDP-8 is running or halted. When the saved document is opened the machine will continue exactly where it was when saved (for now excepting RK05 disks).

If a document is open when exiting PDP-8/IXS, the user will be prompted to save it. Whether or not a document is open, the complete machine state is always saved on exit in the user's Documents folder so that it resumes in exactly the same state when PDP-8/IXS is next run.

A PDP8 document is human-readable XML file and can be examined and edited in any Windows plain-text editor.

Windows 11 is very strict about which exe files are allowed to be associated with a document's file extension. PDP-8/IXS is not currently built for that, so if you try to establish an association using Open With, it won't work. This can be fixed with registry hacking, which any decent LLM can walk you through, but hack the registry at your own risk. If you get `.pdp8` associated with the exe, it will correctly open the document by the usual double-clicking.

OS/8

File→OS-8 boots OS/8 from RK05 unit 0, surface A by reading block 0 into memory at location 0000, waiting for the read to complete, and then doing JMP 0. If no bootable OS/8 exists on the disc, the results are undefined.

An OS/8 disc image exists in the repository, currently named `v3dNew.rk05`. It was derived from the version of OS/8 recovered by the PiDP-8 project, and contains all of the standard OS/8 commands (e.g. PIP, PAL8, ABSLDR, Fortran) and many historically interesting programs. It also contains E8, an Emacs-like editor that runs under OS/8, which can be used on TTY1 by switching it to VT100 mode (more on this below).

PAL8 Listing and Binary Files

A listing generated by PAL8 (or any other PDP-8 assembler) has a variety of uses. It can be used to load object code into memory and to show the source code line being executed during single stepping and for breakpoints and recorded events. Currently the listing file must exist on Windows—the OS/8 file can't yet be directly read. I have a Windows app that can inspect and modify an OS/8-formatted RK05 file, and transfer files between Windows and OS/8. I'll make that app available eventually. The PiDP-8 community may have other ways to transfer files.

Listings have `.lst` or `.LS` extensions. The expectation, not enforced, is that `.lst` files came from non-OS/8 assemblers and `.LS` from PAL8 (of course Windows files are not case sensitive). The listing formats are expected to follow PAL8.

A binary file, extension `.bin` or `.BN`, is a listing with just octal address/contents pairs and no side-by-side source code. This is not OS/8 binary format. A binary file can be loaded into core but source line display is unavailable.

PDP-8 code can also be loaded and run from OS/8 in the usual way.

File menu	Description
Load Listing	Load listing or binary file. Object code is loaded into core, and if listing file load the file into the Listing window.
Read Listing	Load the file into the Listing window. Useful when resuming a saved `.pdp8`.
Split Listing	Split a listing file into three files: `*.bin` (`*.BN`), `*.pal8` (`*.PA`), and `extras.txt`.
Write Binary	Extract the binary portion of a listing and write to a file
Write Source	Extract the source portion of a listing and write to a file
Convert to PAL8	Convert file to upper case, CR/LF line endings. This command will probably be eliminated in subsequent releases.

ASR38/VT100 Terminals

Devices→TTY1 and Devices→TTY2 open terminal windows. TTY1 is the DEC standard console terminal, IOT addresses 03 (keyboard) and 04 (printer). TTY2 uses corresponding IOT addresses 13 and 14.

Each terminal can choose its simulated print speed by setting the $\log_{10}$(character/sec) control, from teletype speed of 10 cps up to 100,000cps.

The reason ASR38 was chosen instead of ASR33, which was actually DEC standard, is to simulate two-color printing, which the MIT RADAR system used. Two-color is not yet implemented.

In ASR38 mode the terminal remembers all printed output, just as paper would. In VT100 mode the window is a screen of 42 lines of 80 characters. The terminal window remembers the contents of the paper or screen as you switch modes.

The only PDP-8 program that actually cares which mode the terminal is in is E8. Switch to VT100 mode before running it from the OS/8 prompt.

The VT100 screen implements the following ANSI escape sequences:

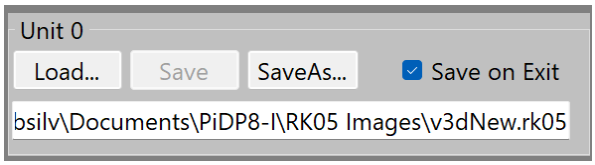
Sequence	Action
ESC [*row;col* H	set cursor position
ESC [2 J	clear screen
ESC [K	clear to end of line
ESC [*top;bot* r	set scrolling window
ESC D	if cursor is at window bottom scroll window up
ESC M	if cursor is at window top scroll window down
ESC c	reset terminal (clear screen, reset scroll window)

The screen will also scroll up if a line feed code (012_8) is received when the cursor is at the bottom of the scroll window.

If the ALT key is pressed in combination with any character, the ESC code (033_8) is transmitted first.

RK05 Disks

Devices→RK05 opens the RK05 window. Controls are shown for all four units. Each looks like this, for example:



The contents of each unit, if any, is loaded from and saved to a Windows file that is compatible with the PiDP-8. The Save button is enabled if any writes to the disk have occurred since it was loaded or saved. Data transfers are by single-cycle breaks.

AF01 A/D Converter

This is the DEC standard device—the *Small Computer Handbook* has the instructions. Channels 1 and 2 read the two sliders on the HRP window.

Priority Interrupt Controller and Mask

The HRP priority interrupt system has 16 interrupt channels, 0 – 15, with 15 being the highest priority, and a 12-bit interrupt mask, where a mask bit is set to disable interrupts on the corresponding channel. Here is the PDP-8/IXS configuration.

Device	IOT Addr	Channel #	Mask Bit
TSVC	57	1	9
interval timer	17	2	10
clock	37	3	11
RK05	74	4	none
integrator	33	8	0
TTY2 printer	14	9	1
TTY1 printer	04	10	2
PT punch	02	11	3
TTY1 keyboard	03	12	4
PT reader	01?	13	5
TTY2 keyboard	13	14	6
supervisor call	57	15	7

Instructions:

ASM	opcode	operation
LDMSK	6561	AC → mask
RDMSK	6562	mask → AC
RDINT	6564	highest priority interrupting channel → AC

Supervisor Call

ASM	opcode	operation
SVC	6571	set SVC interrupt request flag
CLSVC	6573	clear SVC interrupt request flag
XSVC	6572	unknown, ignored
TSVC	6574	toggle TSVC interrupt request flag

Clock and Interval Timer

The interval timer instructions can be OR'd in the usual way.

ASM	opcode	operation
CLK 2	6372	clear clock interrupt request flag
LIT	6171	clear interrupt flag, load interval in ms from AC, start timing
	6172	clear AC
LITC	6173	clear flag, load from AC and start, then 0 → AC
SIT	6174	skip if interrupt request flag is set

Tektronix 611 Storage Scope

Devices→Tektronix 611 opens a sizeable window that emulates the storage scope that came from HRP and was extensively used at MIT. The scope had the normal storage mode, where plotted points stayed put until the entire screen was erased, and non-store mode, where plotted points disappeared after a short interval and the PDP-8 had to continuously redraw the screen. Mode is determined by the checkbox in the lower right corner of the window.

Scope coordinates are 10 bits, with the LSB ignored. The high two bits of the AC are ignored when setting coordinates.

ASM	opcode	operation
SCER	6051	erase scope
SCLX	6052	AC → x coordinate
SCIX	6054	plot point
SCLIX	6056	AC → x and plot
SCSE	6061	skip if scope is erasing
SCLY	6062	AC → y coordinate
SCIY	6064	plot point
SCLIY	6066	AC → y and plot

Other MIT and HRP Hardware

Devices→HRP opens the window on the right.

The appendix lists all of the instructions and data formats. The *RADAR User's Guide* explains the displays and the operation of the controls.

Execution Speed

PDP-8/I cycles are tied to the 1.5 µs core memory read-writeback cycle. Core, like DRAM, is destructive read so every cycle reads and writes back, and read/modify/write cycles like ISZ are just normal cycles with data modified before writeback. Instructions take 1.5 µs for every cycle except for IOT, which uses one core cycle to fetch the instruction but takes 4.25 µs[2].

PDP-8/IXS simulates PDP-8/I timing. Cycles take, on average, 1.5 µs. The extra delay of IOT, called the Pause state, is not simulated. Cycles are run in bursts for various efficiency reasons, so only average cycle time is accurate. This is not noticeable to the human observer, and in principle not noticeable to running code.

Options→10x makes the simulated cycle time 0.15 µs, i.e. ten times faster. RK05 transfers also speed up 10x, but I/O operations like printing speed do not.

The main front panel window (see above picture) shows the current target cycle time (1.50 or 0.15), the actual average time the host CPU needs to run a cycle, and the average percent of time that the thread on the host CPU is active. These averages are computed by a low-pass filter with a 1-second time constant.

I/O instruction cycles, such as TTY output, and event recording can run much slower than a normal cycle due to for example lots of GUI screen updating. These increase the workload on the host thread but do not change the cycle frequency as long as the host can keep up. PDP-8/IXS tries to keep the host thread workload under 100%, but there are cases where it can't. Above 100% just means cycle timing is no longer accurate.

Hardware Breakpoint

The hardware breakpoint is controlled by the switch register. The high-order three bits (also used for Data Field entry) control the break mode:

[2] *Small Computer Handbook*, 1970 Edition, page 35

Bit	State	Effect
0	0	break on core field and address matching SR bits 3 - 17
	1	break on core data matching SR bits 6 - 17
1	0	don't break on core writes
	1	break on matching core writes
2	0	don't break on core reads
	1	break on matching core reads

If bits 1 and 2 are 0 (off), the breakpoint is disabled. If they are both set, break will occur on a matching read or write (i.e. any access). The one- and three-cycle data breaks are also monitored and can cause a break. Note that addresses are 15 bits, SR 3-17.

The break occurs on a machine cycle boundary, which can be in the middle of an instruction, as if the Single Step switch was set momentarily at just the right time.

Note that on the actual PDP-8/I, every core cycle was a read/modify/write due to the nature of magnetic core. The hardware breakpoint is based on a programmer's view of what is a read or write (or both, e.g. ISZ).

When a break occurs, if event recording is disabled the CPU will halt with interrupts suppressed. Otherwise an event will be recorded and execution continues.

Event Recording

When certain events occur during instruction execution, and recording is enabled by the Analyze→Record checkbox, system state information can be recorded. The events are

- A JMS instruction
- A JMP indirect instruction
- An SVC instruction
- An interrupt
- Breakpoint match

The Analyze→Record Only Breaks checkbox ignores all events except breakpoint match. The state recorded includes

- Complete CPU state
- Switch register
- State of all I/O flags
- Cycle count and realtime value

When the CPU isn't running you can examine the last 1000 recorded events using the commands in the Analyze menu. The index of the event record being examined is displayed in the main front panel window. A value of n means n events before the last one recorded. Events are in a ring buffer. An event is displayed by setting the main front panel window to show CPU state, SR, cycle count, real time, and the IRQ consequences of the saved I/O flags and interrupt mask. If a listing was loaded the source code line corresponding to the event is highlighted in

the Listing window. It shows the instruction being executed, not the next one that the disassembler shows.

Appendix—HRP & MIT IOT Instructions

HRP provided a switch to choose one of two radars, which they called side and front (of a C-130). MIT used WR66 (10 cm klystron) and WR73 (5 cm magnetron). WR66 azimuth and elevation readings came from 14-bit synchro-to-digital converters producing binary angles. WR73 AZ and EL come from binary-coded-decimal angles in tenths degree. For the DNS instructions read WR73 BCD angles (0 is tenths digit, 1 unit, 2 tens, 3 hundreds):

azimuth		elevation	
DNS 2	DNS 7	DNS 7	DNS 3
332222111100	00----------	----------33	222211110000

Integrator Control Words:

bit	ICW1	ICW2
0	Skip bins	start?
1		WR66/WR73
2		TNP
3		
4		
5		# pulses
6		
7		
8		
9	2.0 nmi bins	
10	1.0 nmi bins	
11	0.5 nmi bins	

External Switch Register (ESR):

bit	ESR 1	ESR 2
0		A-Scope display dropdown
1		
2		
3	Zero button	
4		
5	Next Screen button	
6		
7	Radar Sel button toggles	
8	Enable send to TI-980	
9	Run checkbox	
10		
11		

WR66 Servo Interface. These values are displayed on the HRP window but don't yet actually move the antenna.

<table>
<tr><th>bit</th><th>WR66 High</th><th>WR66 Low</th></tr>
<tr><td>0</td><td rowspan="4">high bits of elevation velocity</td><td rowspan="4">low bits of elevation velocity</td></tr>
<tr><td>1</td></tr>
<tr><td>2</td></tr>
<tr><td>3</td></tr>
<tr><td>4</td><td></td><td rowspan="8">azimuth velocity</td></tr>
<tr><td>5</td><td></td></tr>
<tr><td>6</td><td></td></tr>
<tr><td>7</td><td></td></tr>
<tr><td>8</td><td></td></tr>
<tr><td>9</td><td rowspan="2">control bits (ignored)</td></tr>
<tr><td>10</td></tr>
<tr><td>11</td><td></td></tr>
</table>

Here are the instructions:

ASM	opcode	operation
INA 1	6111	latch S/D (ignored)
INA 2	6112	low 12 bits of WR66 AZ → AC
INA 3	6113	low 12 bits of WR66 EL → AC
INA 7	6117	high 2 bits of WR66 EL (bits 8-9) and AZ (bits 10-11) → AC
DNS 2	6122	high bits of WR73 AZ → AC
DNS 3	6123	low bits of WR73 AZ → AC
DNS 7	6127	low bits of WR73 AZ and high bits of EL → AC
TIR	6161	ignored, originally read from TI-980
TIW	6162	skips, originally write to TI-980
FRE 4	6314	release S/D (ignored)
SRA 2	6322	load integrator control word 1 (ICW1)
SRA 6	6326	load integrator control word 2 (ICW2)
RVI 2	6332	clear integrator interrupt request flag
RVI 4	6334	read next range bin from integrator (currently reads synthetic values)
INB 1	6351	latch/release WR73 angles (ignored)
ESR 2	6362	ESR select = 0010: ESR2 → AC ESR select = 4000: ESR1 → AC
ESR 5	6365	AC → ESR select
TL2 1	6411	skip if Spy button has been pressed since last TL2 1
TL2 2	6412	0 → AC
TL2 4	6414	AC |= Spy digit from dropdown list
TL2 6	6416	Spy digit → AC
PEL73	6422	AC → WR73 elevation
INI	6441	AC → WR73 azimuth velocity
OSW 1	6451	AC → OSW, bit 7 is WR66 (0) WR73 (1), others unknown
OSW 2	6452	AC → 2-digit 7-segment display (Idle #), bits 4-7, 8-11 BCD digits
RH66	6514	0 → AC (unused by RADAR, original purpose unknown)
RL66	6515	0 → AC (unused by RADAR, original purpose unknown)
WL66	6516	AC → low bits of WR66 servo
WH66	6517	AC → high bits of WR66 servo